

1. Creating and Renaming Files/Directories

Create a directory named test_dir using mkdir.

Inside test_dir, create an empty file called example.txt

Rename example.txt to renamed_example.txt using mv

```
manoj@Manojs-MacBook-Pro LINUX % cd Assignment
manoj@Manojs-MacBook-Pro Assignment % mkdir test_dir
manoj@Manojs-MacBook-Pro Assignment % ls
test_dir
manoj@Manojs-MacBook-Pro Assignment % cd test_dir
manoj@Manojs-MacBook-Pro test_dir % touch example.txt
manoj@Manojs-MacBook-Pro test_dir % ls
example.txt
manoj@Manojs-MacBook-Pro test_dir % mv example.txt renamed_example.txt
manoj@Manojs-MacBook-Pro test_dir % ls
renamed_example.txt
manoj@Manojs-MacBook-Pro test_dir % █
```

2. Viewing File Contents

Use cat to display the contents of /etc/passwd

Display only the first 5 lines of /etc/passwd using head.

Display only the last 5 lines of /etc/passwd using tail.

```
manoj@Manojs-MacBook-Pro ~ % cat /etc/passwd
# User Database
#
# Note that this file is consulted directly only when the system is running
# (i.e. login time); at other times this information is obtained by
# fetching from the database by the 'getent' command, by running 'gpasswd'
# using the 'newgrp' command, or by the 'chsh' command. The format of the
# file is as follows:
#
# Username:Password:UID:GID:Geometric Name:Room:Shell:
#-----
root:x:0:0:System Administrator:/var/root:/bin/sh
daemon:x:1:1:System Services:/var/empty:/usr/bin/false
bin:x:2:2:bin:/var/empty:/usr/bin/false
manoj@Manojs-MacBook-Pro ~ % head -5 /etc/passwd
root:x:0:0:System Administrator:/var/root:/bin/sh
daemon:x:1:1:System Services:/var/empty:/usr/bin/false
bin:x:2:2:bin:/var/empty:/usr/bin/false
manoj@Manojs-MacBook-Pro ~ % tail -5 /etc/passwd
manoj@Manojs-MacBook-Pro ~ % █
```

3. Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd

```
manoj@Manojs-MacBook-Pro ~ % grep -n root /etc/passwd
12:root:x:0:0:System Administrator:/var/root:/bin/sh
13:daemon:x:1:1:System Services:/var/empty:/usr/bin/false
62:_cvmroots:x:212:212:CVMS Root:/var/empty:/usr/bin/false
manoj@Manojs-MacBook-Pro ~ % █
```

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip

Unzip test_dir.zip into a new directory named unzipped_dir.

```
manoj@Manojs-MacBook-Pro Assignment % zip test_dir.zip test_dir
adding: test_dir/ (stored 0%)
manoj@Manojs-MacBook-Pro Assignment % ls
test_dir  test_dir.zip
manoj@Manojs-MacBook-Pro Assignment % █
```

```
manoj@Manojs-MacBook-Pro Assignment % unzip test_dir.zip -d unzipped_dir
Archive: test_dir.zip
creating: unzipped_dir/test_dir
manoj@Manojs-MacBook-Pro Assignment % ls
ASSIGNMENT.pdf  SECURITY.md  unzipped_dir
LINUX ASSIGNMENT.pdf  test_dir  test_dir.zip
secure.txt  test_dir.zip
manoj@Manojs-MacBook-Pro Assignment % █
```

5. Downloading Files

Use wget to download a file from a URL (e.g., <https://example.com/sample.txt>).

```
manoj@Manojs-MacBook-Pro Assignment % wget https://raw.githubusercontent.com/molibot/molibot/refs/heads/main/SECURITY.md
--2026-01-30 00:06:17-- https://raw.githubusercontent.com/molibot/molibot/refs/heads/main/SECURITY.md
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 2606:58c0:8008::154, 2606:58c0:8008::154, 2606:58c0:8008::154, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|2606:58c0:8008::154|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1619 (1.6K) [text/plain]
Saving to: 'SECURITY.md'

SECURITY.md                                     100%[=====]
2026-01-30 00:06:17 (5.15 MB/s) - 'SECURITY.md' saved [1619/1619]

manoj@Manojs-MacBook-Pro Assignment % ls
SECURITY.md  test_dir  test_dir.zip
manoj@Manojs-MacBook-Pro Assignment % █
```

6. Changing Permissions

Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

```
manoj@Manojs-MacBook-Pro Assignment % touch secure.txt
manoj@Manojs-MacBook-Pro Assignment % chmod +r secure.txt
manoj@Manojs-MacBook-Pro Assignment % ls -la secure.txt
-rw-r--r-- 1 manoj staff 0 30 Jan 00:06 secure.txt
manoj@Manojs-MacBook-Pro Assignment % chmod +r -w secure.txt
chmod: -w: No such file or directory
manoj@Manojs-MacBook-Pro Assignment % chmod +r-w secure.txt
manoj@Manojs-MacBook-Pro Assignment % ls -la
total 16
drwxr-xr-x 3 manoj staff 192 30 Jan 00:06 .
drwxr-xr-x 3 manoj staff 96 29 Jan 23:53 ..
-r--r--r-- 1 manoj staff 0 30 Jan 00:06 secure.txt
-rw-r--r-- 1 manoj staff 1619 30 Jan 00:06 SECURITY.md
drwxr-xr-x 3 manoj staff 96 29 Jan 23:54 test_dir
-rw-r--r-- 1 manoj staff 168 30 Jan 00:01 test_dir.zip
manoj@Manojs-MacBook-Pro Assignment % ls -la secure.txt
-r--r--r-- 1 manoj staff 0 30 Jan 00:06 secure.txt
manoj@Manojs-MacBook-Pro Assignment % █
```

7. Working with Environment Variables

Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

```
manoj@Manojs-MacBook-Pro Assignment % export MY_VAR = 'Hello, Linux'
zsh: bad assignment
manoj@Manojs-MacBook-Pro Assignment % export MY_VAR='Hello, Linux'
manoj@Manojs-MacBook-Pro Assignment % echo $MY_VAR
Hello, Linux
manoj@Manojs-MacBook-Pro Assignment % █
```

Submission Guidelines -: Attach Screenshots or command along with explanation and submit in doc(google doc or microsoft doc) format also attach github repo link

